



KINGDOM OF SAUDI ARABIA | JAZAN UNIVERSITY
COLLEGE OF ENGINEERING AND COMPUTER SCIENCE

ASSIGNMENT 2 – 2025/2026 (Semester 1)

Course with code	Data Mining (COMP-454)	Section	2312
Type of Assignment	Group Assignment (Theory)	Marks	15 (10 written & 5 viva)
Course Instructor	Dr. Shatha Ali A. Hakami	Signature	
Date of Announcement	13-04-2026	Deadline	02-05-2026
Group 3			
Students Names	Marwh zed	IDs Numbers	202306845
	Jamela jaber ali		202009375
	Rafa Jerebi		202303417

ASSIGNMENT SUBMISSION INSTRUCTIONS

You must **KEEP THIS PAGE** in your submitted assignment.

You must write your group members **Names** and **IDs** in the corresponding fields above.

You must answer **the questions assigned specifically to your group**, failing to answer your assigned version of questions will result in **ZERO** marks out of 15.

Copying the answers from your classmate is considered cheating and will result in a **ZERO** mark.

Write your answers in **English** language only.

Write your answers in this file, then save it as a **PDF** and submit it via **Blackboard** only.

Submit this assignment on time as stated above; **LATE ASSIGNMENTS ARE NOT ACCEPTABLE.**

Only soft-copy assignments will be accepted (no hard-copy).



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ASSIGNMENT PROBLEM STATEMENT

Education Dataset

This dataset contains information about students' academic performance and habits, including study time, attendance, assignments, sleep, and internet usage. Predict whether a student will pass and analyze the impact of study behaviors on performance.

Tasks: using the provided dataset, apply the following:

marks

Task 1: Data Understanding	Explore and describe the dataset by identifying its structure, types of attributes (e.g., numerical, categorical, binary), and the overall purpose of the data. You should clearly define the target variable and explain the real-world context of the dataset.	1
Task 2: Data Preprocessing	Prepare the dataset for analysis by handling missing values, detecting and treating noise or inconsistencies, and applying appropriate data transformation techniques (e.g., normalization or encoding). All preprocessing steps must be clearly justified.	1.5
Task 3: Exploratory Data Analysis (EDA)	Analyze the dataset using statistical summaries and visualizations (e.g., histograms, boxplots, scatter plots) to understand data distribution, relationships between variables, and key patterns or trends.	1.5
Task 4: Pattern Mining	Apply frequent pattern mining or association rule mining techniques. You must define suitable minimum support and confidence thresholds, extract meaningful rules, and interpret their significance within the dataset context.	1.5
Task 5: Classification	Build a classification model using at least one algorithm (e.g., Decision Tree, naïve Bayes, or k-NN). You should train the model, evaluate its performance (e.g., accuracy), and explain the results and their implications.	1.5
Task 6: Cluster Analysis	Apply at least one clustering algorithm, such as K-Means, Hierarchical clustering, or DBSCAN. You should select appropriate parameters (e.g., number of clusters K for K-Means) and justify the choice of parameters. Visualize the clustering results.	1.5
Task 7: Outlier Detection	Identify outliers in the dataset using appropriate methods (e.g., statistical measures or visualization). You must explain why these data points are considered outliers and discuss their possible real-world meaning.	1.5
Task 8: The Presentation	Your group should present all seven tasks in 7 PowerPoint slides , with each task presented on a separate slide.	5
Total		15



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Start Writing Your Answer Here.

1. Data Understanding (Task 1)

The dataset contains information about students' academic performance and study habits.

It includes the following attributes:

- StudyHours (Numerical)
- Attendance (Numerical)
- Assignments (Numerical)
- InternetUsage (Numerical)
- GroupStudy (Binary)
- SleepHours (Numerical)
- Pass (Categorical → Target Variable)

The target variable is "Pass", which indicates whether a student will pass or fail.

This dataset can be used in real-world scenarios to predict student success and improve academic performance by analyzing study behaviors.

1. Import Libraries & Load Dataset

```
# Import required libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_csv('/content/G3_education.csv')

# Display first rows
df.head()
```

```
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datet
return datetime.utcnow().replace(tzinfo=utc)
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return datetime.utcnow().replace(tzinfo=utc)
```

StudentID	StudyHours	Attendance	Assignments	InternetUsage	GroupStudy	SleepHours	Pass	
0	1.0	10.0	98.0	NaN	4.0	1.0	7.0	No
1	2.0	2.0	51.0	0.0	NaN	0.0	7.0	No
2	3.0	11.0	88.0	0.0	5.0	0.0	5.0	Yes
3	4.0	4.0	98.0	9.0	8.0	1.0	9.0	Yes
4	5.0	2.0	70.0	4.0	9.0	0.0	6.0	Yes

```
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```



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2. Data Understanding

```
[ ] # Display dataset structure
df.info()

# Statistical summary
df.describe()

# Display column names
df.columns

▼ <class 'pandas.core.frame.DataFrame'>
RangeIndex: 120 entries, 0 to 119
Data columns (total 8 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   StudentID    117 non-null    float64
1   StudyHours   110 non-null    float64
2   Attendance   113 non-null    float64
3   Assignments  112 non-null    float64
4   InternetUsage 115 non-null    float64
5   GroupStudy   114 non-null    float64
6   SleepHours   116 non-null    float64
7   Pass         116 non-null    object
dtypes: float64(7), object(1)
memory usage: 7.6+ KB
/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC.
return datetime.datetime.utcnow().replace(tzinfo=utc)
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return datetime.datetime.utcnow().replace(tzinfo=utc)
Index(['StudentID', 'StudyHours', 'Attendance', 'Assignments', 'InternetUsage',
      'GroupStudy', 'SleepHours', 'Pass'],
      dtype='object')
```

2. Data Preprocessing (Task 2)

Data preprocessing was performed to clean and prepare the dataset.

- Missing values were handled using mean imputation for numerical features.
- The target variable "Pass" was encoded into binary values (0 = No, 1 = Yes).
- Data consistency was ensured by missing values in the target column were filled using the mode.

These steps improve data quality and make it suitable for machine learning models.



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1
Ds

```
# Before
missing_before = df.isnull().sum()

# Fill numerical with mean
df.fillna(df.mean(numeric_only=True), inplace=True)

# Fill categorical (Pass) with mode
df['Pass'] = df['Pass'].fillna(df['Pass'].mode()[0])

# After
missing_after = df.isnull().sum()

missing_df = pd.DataFrame({
    'Before': missing_before,
    'After': missing_after
})

print(missing_df)
```

```
***          Before  After
StudentID         3      0
StudyHours        10      0
Attendance         7      0
Assignments        8      0
InternetUsage      5      0
GroupStudy         6      0
SleepHours         4      0
Pass              4      0
```

Pass unique values: [0 1]

/usr/local/lib/python3.12/dist-packages/jupyter_client/session.py:203: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use timezone-aware objects to represent datetimes in UTC: datetime.datetime.now(datetime.UTC).
return datetime.datetime.utcnow().replace(tzinfo=utc)

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return datetime.datetime.utcnow().replace(tzinfo=utc)

	StudyHours	Attendance	Assignments	InternetUsage	GroupStudy	SleepHours	Pass	Cluster
0	10.0	98.0	4.508929	4.0	1.0	7.0	0	1
1	2.0	51.0	0.000000	5.0	0.0	7.0	0	0
2	11.0	88.0	0.000000	5.0	0.0	5.0	1	1
3	4.0	98.0	9.000000	8.0	1.0	9.0	1	1
4	2.0	70.0	4.000000	9.0	0.0	6.0	1	2

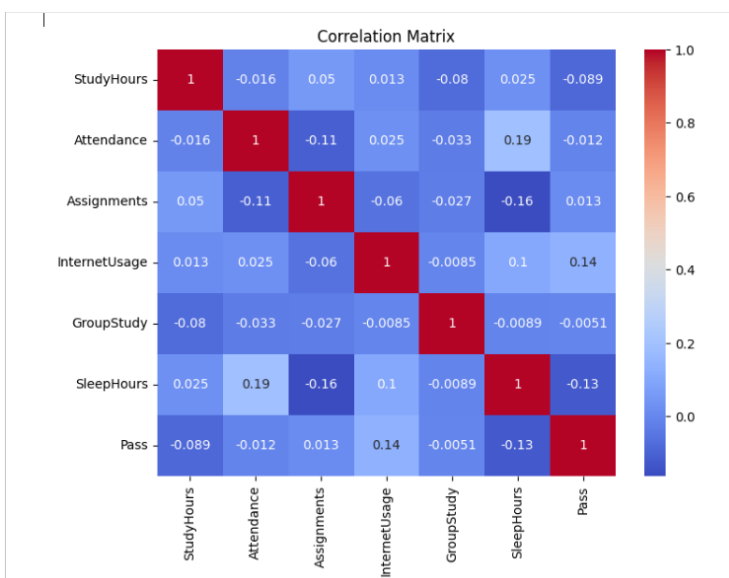
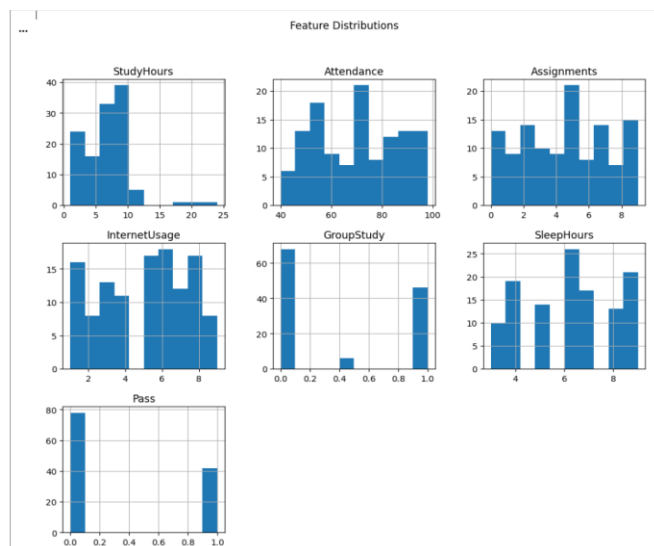


3. EDA (Task 3)

Exploratory Data Analysis (EDA) was conducted to understand the data distribution and relationships.

- Histograms were used to visualize feature distributions.
- A correlation heatmap was used to identify relationships between variables.

The analysis showed that features such as attendance and study hours have noticeable impact on student performance.





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4. Pattern Mining (Task 4)

Association rule mining was applied using the Apriori algorithm.

Minimum support was set to 0.3 and minimum confidence to 0.6 to extract meaningful rules.

The dataset was converted into binary format before applying the algorithm.

Example rule:

Students with higher assignment scores are more likely to achieve better academic outcomes

These rules help uncover hidden relationships between student behaviors and performance.

```
return datetime.utcnow().replace(tzinfo=utc)
```

	antecedents	consequents	support	confidence	lift
0	(StudyHours)	(Cluster)	0.316667	0.644068	0.990874
1	(Attendance)	(Cluster)	0.466667	1.000000	1.538462
2	(Cluster)	(Attendance)	0.466667	0.717949	1.538462
3	(InternetUsage)	(Cluster)	0.308333	0.672727	1.034965
4	(SleepHours)	(Cluster)	0.308333	0.725490	1.116139

5. Classification (Task 5)

The Decision Tree model achieved an accuracy of 67%, indicating a moderate performance in predicting student outcomes.

The model performed better in predicting students who did not pass (Class 0) with a higher F1-score of 0.75, compared to students who passed (Class 1) with an F1-score of 0.50.

This suggests that the model is slightly biased toward the majority class and may require further tuning or more balanced data for improved performance.

```
return datetime.utcnow().replace(tzinfo=utc)  
Accuracy: 0.6666666666666666
```

Classification Report:

	precision	recall	f1-score	support
0	0.71	0.80	0.75	15
1	0.57	0.44	0.50	9
accuracy			0.67	24
macro avg	0.64	0.62	0.62	24
weighted avg	0.66	0.67	0.66	24



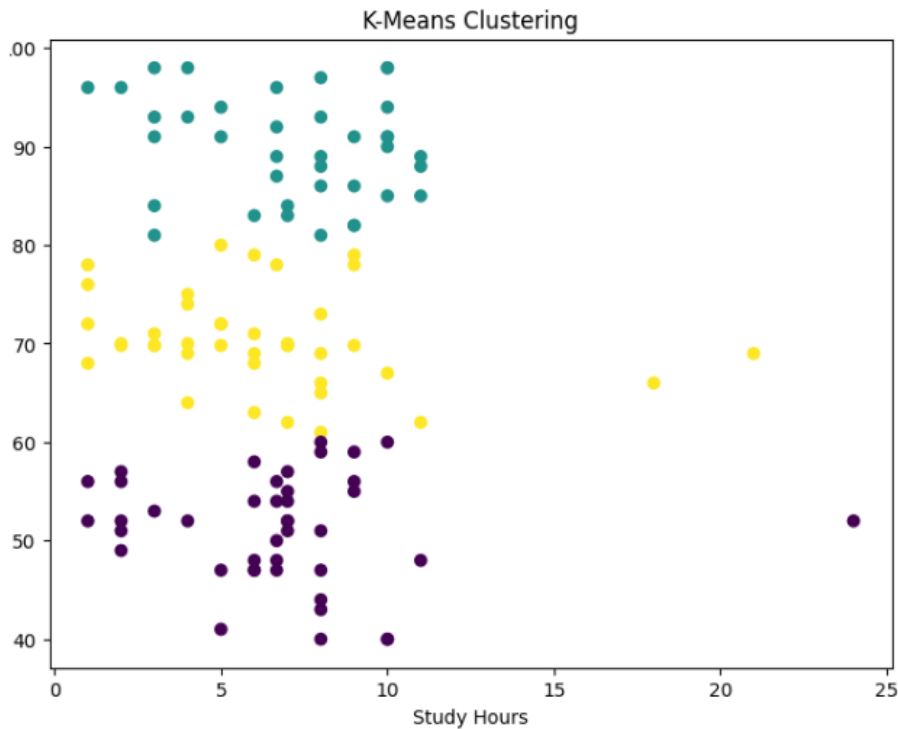
6. Clustering (Task 6)

The K-Means clustering algorithm was applied with $K=3$ to group students based on their study habits.

The results show three distinct clusters:

- Cluster 1 represents students with low study hours and lower performance.
- Cluster 2 represents students with moderate study habits and average performance.
- Cluster 3 represents students with high study hours and better academic performance.

This indicates a strong relationship between study behavior and student success.





7. Outlier Detection (Task 7)

Boxplots were used to identify potential outliers in the dataset.

The analysis shows that most features follow a relatively normal distribution without significant extreme values.

However, the *StudyHours* feature contains a few outliers, indicating that some students spend unusually high amounts of time studying compared to others.

These values may represent highly dedicated students or exceptional study behavior rather than errors.

Therefore, the outliers were retained as they provide meaningful insights into variations in student performance.

